

3D-LINES



SYNOPSIS

Equation of a line:

➤ General Form (Unsymmetrical form) of a line:

The intersection of two plane s is a line. The equations

$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$ represents a line.

i) Equation to the X-axis is $y = 0, z = 0$

ii) Equation to the Y-axis is $x = 0, z = 0$

iii) Equation to the Z-axis is $x = 0, y = 0$

iv) Equation of the line parallel to x-axis is $y = p, z = q, p, q \in \mathbb{R}$

v) Equation of the line parallel to y-axis is $x = h, z = q, h, q \in \mathbb{R}$

vi) Equation of the line parallel to z-axis is $x = h, y = p, h, p \in \mathbb{R}$

Symmetrical form of a line:

➤ The equation of the line passing through the point (x_1, y_1, z_1) and having d.c's (l, m, n) is

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

Proof: Let L be the line passing through (x_1, y_1, z_1) with d.c's (l, m, n) , Let $A = (x_1, y_1, z_1)$

Let $P(x, y, z)$ be any point on L Let $AP = r$

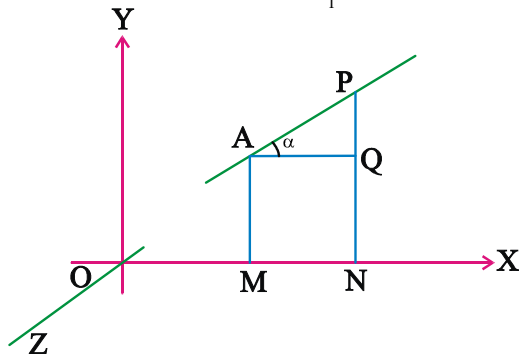
Let α be the angle which L makes with x-axis

Let M, N be the projections of A, P on x-axis respectively.

Let Q be the projections of A on $\overline{PN}$,

$OM = x_1, ON = x$

$MN = ON - OM = x - x_1$



Also $MN =$ projection of AP on x-axis

$$MN = r \cdot \cos \alpha$$

$$x - x_1 = r \cdot l$$

$$\frac{x - x_1}{l} = r$$

$$\left[\begin{array}{l} \text{From } \triangle APQ \\ \cos \alpha = \frac{AQ}{AP} \\ AQ = AP \cdot \cos \alpha \\ MN = r \cdot \cos \alpha \end{array} \right]$$

Similarly we can prove $\frac{y - y_1}{m} = \frac{z - z_1}{n} = r$

Hence the equation of the line L is

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

i) The equation of a line passing through the point (x_1, y_1, z_1) and having d.r's (a, b, c) is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

ii) The equation of the line passing through two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

Vector form of a line:

➤ Cartesian equation of a line passing through the point (x_1, y_1, z_1) and having d.r's (a, b, c) is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

$$\text{Let } \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = \lambda$$

$$x - x_1 = a\lambda \Rightarrow x = x_1 + a\lambda$$

$$y - y_1 = b\lambda \Rightarrow y = y_1 + b\lambda$$

$$z - z_1 = c\lambda \Rightarrow z = z_1 + c\lambda, \text{ Now,}$$

$$x\vec{i} + y\vec{j} + z\vec{k} = x_1\vec{i} + \lambda a\vec{i} + y_1\vec{j} + \lambda b\vec{j} + z_1\vec{k} + \lambda c\vec{k}$$

$$\vec{r} = (x_1\vec{i} + y_1\vec{j} + z_1\vec{k}) + \lambda(a\vec{i} + b\vec{j} + c\vec{k})$$

Which is the vector equation of the line passing through (x_1, y_1, z_1) and having d.r's (a, b, c)

(or) vector equation of the line passing through (x_1, y_1, z_1) and parallel to the vector $a\vec{i} + b\vec{j} + c\vec{k}$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.

W.E-1: The cartesian equation and vector equation of the line passing through the points $(-1, 0, 2)$ and $(3, 4, 6)$

Sol: Let $(x_1, y_1, z_1) = (-1, 0, 2)$

$(x_2, y_2, z_2) = (3, 4, 6)$

D.r's of the line $= (x_2 - x_1, y_2 - y_1, z_2 - z_1)$

let $(a, b, c) = (4, 4, 4)$

Cartesian equation of the line is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

$$\Rightarrow \frac{x + 1}{4} = \frac{y - 0}{4} = \frac{z - 2}{4}$$

$$\text{Let } \frac{x + 1}{4} = \frac{y - 0}{4} = \frac{z - 2}{4} = \lambda$$

$$x + 1 = 4\lambda \Rightarrow x = -1 + 4\lambda$$

$$y = 4\lambda$$

$$z - 2 = 4\lambda \Rightarrow z = 2 + 4\lambda$$

$$x\bar{i} + y\bar{j} + z\bar{k} = -\bar{i} + 4\lambda\bar{i} + 4\lambda\bar{j} + 2\bar{k} + 4\lambda\bar{k}$$

$$\bar{r} = (-\bar{i} + 2\bar{k}) + \lambda(4\bar{i} + 4\bar{j} + 4\bar{k})$$

$$\text{Where } \bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$$

Which is the vector equation of the line.

Conversion of non-symmetrical form to symmetrical form:

➤ Let the equation of the line in non-symmetrical form be

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$$

To find the equation of the line in symmetrical form, we must know (i) its d.r's (ii) coordinates of any point on it.

i) To find the d.r's of the line:

Let l, m, n be the d.r's of the line.

Since the line lies in both the planes, it must be perpendicular to normals of both planes.

$$\text{So, } a_1l + b_1m + c_1n = 0$$

$$a_2l + b_2m + c_2n = 0$$

From these equations proportional values of l, m, n found by cross multiplication method

$$\begin{array}{ccccc} l & m & n \\ b_1 & c_1 & a_1 & b_1 \\ b_2 & c_2 & a_2 & b_2 \end{array}$$

$$\Rightarrow \frac{l}{b_1c_2 - b_2c_1} = \frac{m}{c_1a_2 - c_2a_1} = \frac{n}{a_1b_2 - a_2b_1}$$

ii) To find a point on the line:

At least one of the d.r's must be non-zero.

$$\text{Let } a_1b_2 - a_2b_1 \neq 0$$

The line cannot be parallel to xy-plane.

Let it intersect the xy-plane in $(x_1, y_1, 0)$

$$\text{then } a_1x_1 + b_1y_1 + d_1 = 0$$

$$\text{and } a_2x_1 + b_2y_1 + d_2 = 0$$

By solving these equations we get the point $(x_1, y_1, 0)$ on the line.

Hence the equation of the line in symmetric form

$$\text{is } \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - 0}{n}$$

Note: If $l \neq 0$, take a point on yz-plane as $(0, y_1, z_1)$

and if $m \neq 0$ take a point on xz-plane as $(x_1, 0, z_1)$.

W.E-2: The equation of the line

$3x + 2y - z - 4 = 0 = 4x + y - 2z + 3$ in symmetrical form is

Sol: Let (a, b, c) be the d.r's of the line then

$$3a + 2b - c = 0, 4a + b - 2c = 0$$

By cross multiplication method

$$\frac{a}{-3} = \frac{b}{2} = \frac{c}{-5}$$

d.r's of the line are $(-3, 2, -5)$

Since $c \neq 0$, the line not parallel to xy-plane.

Let $(x_1, y_1, 0)$ be a point on the line

$$3x_1 + 2y_1 - 4 = 0, 4x_1 + y_1 + 3 = 0$$

By solving these equations we get

$$x_1 = -2, y_1 = 5$$

A point on the line is $(-2, 5, 0)$

Hence the equation of the line in symmetrical form

$$\frac{x + 2}{-3} = \frac{y - 5}{2} = \frac{z - 0}{-5}$$

Parametric form:

➤ The parametric equations of the line passing through the point $P(x_1, y_1, z_1)$ and having d.c's

$$(l, m, n) \text{ are } x = x_1 + lr, y = y_1 + mr,$$

$$z = z_1 + nr \text{ Where } r = \text{OP}$$

Remark: The coordinates of a point on the line whose d.c's are (l, m, n) which is at a distance of 'r' units from the point (x_1, y_1, z_1) are

$$(x_1 \pm lr, y_1 \pm mr, z_1 \pm nr)$$

W.E-3. The coordinates of a point on the line

$\frac{x-1}{2} = \frac{y+1}{-3} = z$ at a distance of $4\sqrt{14}$ from the point $(1, -1, 0)$ nearer to the origin are

Sol. Let $A = (1, -1, 0)$

$$\frac{x-1}{2} = \frac{y+1}{-3} = z = t$$

The coordinates of any point P on the given line are $(2t+1, -3t-1, t)$

$$AP = 4\sqrt{14}$$

$$(2t)^2 + (-3t)^2 + (t)^2 = (4\sqrt{14})^2 \Rightarrow t = \pm 4$$

So the coordinates of the required point are $(9, -13, 4)$ and $(-7, 11, -4)$

Out of which nearer to the origin is $(-7, 11, -4)$

Angle between two lines:

➤ If θ is the angle between the lines given by

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \text{ and}$$

$$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \text{ then}$$

$$\cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}}$$

i) a) If the lines are parallel then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

b) If the lines are $\perp$ then $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

ii) If ' θ ' is the acute angle between the line

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} \text{ and the plane}$$

$$ax + by + cz + d = 0 \text{ then}$$

$$\sin \theta = \frac{|al + bm + cn|}{\sqrt{a^2 + b^2 + c^2} \sqrt{l^2 + m^2 + n^2}}$$

iii) If the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ is parallel

to the plane $ax + by + cz + d = 0$ then $al + bm + cn = 0$ (Normal to the plane is perpendicular to the line)

iv) If the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ perpendicular to the plane $ax + by + cz + d = 0$ then

$$\frac{a}{l} = \frac{b}{m} = \frac{c}{n}.$$

v) d.c.'s of the line which makes equal angles with

coordinate axes are $\pm \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$ and the

d.r.'s of the line are $(1, 1, 1)$.

Coplanar lines:

➤ Two lines are said to be coplanar if they are either parallel or intersect.

Non-Coplanar Lines:

➤ Two lines are said to be non coplanar or skew lines if they are neither parallel nor intersecting.

Condition for two lines to be coplanar:

➤ The line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ lies in the plane

$$ax + by + cz + d = 0 \text{ if}$$

$$ax_1 + by_1 + cz_1 + d = 0, al + bm + cn = 0.$$

Proof: let $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} \dots \dots \dots (1)$

$$ax + by + cz + d = 0 \dots \dots \dots (2)$$

if the line (1) lies in the plane (2) then for all values 't' the point

$p(x_1 + lt, y_1 + mt, z_1 + nt)$ will lie in the plane (2)

$$\text{ie, } a(x_1 + lt) + b(y_1 + mt) + c(z_1 + nt) + d = 0$$

$$\Rightarrow t(al + bm + cn) + (ax_1 + by_1 + cz_1 + d) = 0$$

it is true for all values of 't'

hence we must have $al + bm + cn = 0$ and

$$ax_1 + by_1 + cz_1 + d = 0$$

which are the required conditions.

Remark: The lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$,

$$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \text{ are coplanar}$$

$$\Leftrightarrow \begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

W.E-4. The lines

$$\frac{x-a+d}{\alpha-\delta} = \frac{y-a}{\alpha} = \frac{z-a-d}{\alpha+\delta}$$

$$\text{and } \frac{x-b+c}{\beta-\gamma} = \frac{y-b}{\beta} = \frac{z-b-c}{\beta+\gamma} \text{ are}$$

Sol: $(x_1, y_1, z_1) = (a-d, a, a+d)$

$$(x_2, y_2, z_2) = (b-c, b, b+c)$$

$$(a_1, b_1, c_1) = (\alpha-\delta, \alpha, \alpha+\delta)$$

$$(a_2, b_2, c_2) = (\beta-\gamma, \beta, \beta+\gamma)$$

$$\text{now } \begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

$$= \begin{vmatrix} a-d-b+c & a-b & a+d-b-c \\ \alpha-\delta & \alpha & \alpha+\delta \\ \beta-\gamma & \beta & \beta+\gamma \end{vmatrix}$$

$$c_1 \rightarrow c_1 + c_2 + c_3$$

$$= \begin{vmatrix} 3(a-b) & a-b & a+d-b-c \\ 3\alpha & \alpha & \alpha+\delta \\ 3\beta & \beta & \beta+\gamma \end{vmatrix} = 0$$

Hence given lines are coplanar

. Equation of a plane containing lines:

➤ The equation of the plane containing the lines

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1},$$

$$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \text{ is}$$

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$(\text{or}) \begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

W.E-5 The value of K, if the lines

$$\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-k}{5} \text{ and}$$

$$\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5} \text{ are coplanar}$$

Sol: points on the given

$$\text{lines } (x_1, y_1, z_1) = (-3, 1, k)$$

$$(x_2, y_2, z_2) = (-1, 2, 5)$$

$$\text{D.r's of given lines } (a_1, b_1, c_1) = (-3, 1, 5)$$

$$(a_2, b_2, c_2) = (-1, 2, 5)$$

Equation of the plane is

$$\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} -2 & -1 & k-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = 0$$

$$\Rightarrow K = 5$$

➤ If the lines $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$,

$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$ are coplanar then

$$\frac{a_1x_1 + b_1y_1 + c_1z_1 + d_1}{a_1l + b_1m + c_1n} = \frac{a_2x_1 + b_2y_1 + c_2z_1 + d_2}{a_2l + b_2m + c_2n}$$

Skew lines:

- Two straight lines are said to be skew lines if they are neither parallel nor intersecting, i.e. the lines which do not lie in a plane.

Shortest distance:

- If L_1 and L_2 are skew lines then there is one and only one line perpendicular to both of the lines L_1 and L_2 which is called the line of shortest distance. If PQ is the line of shortest distance then the distance between P and Q is called distance between the given skew lines.

i) The shortest distance between the skew lines

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1, \vec{r} = \vec{a}_2 + \mu \vec{b}_2 \text{ is}$$

$$\frac{|(\vec{a}_1 - \vec{a}_2) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} \text{ (or)} \frac{|\begin{vmatrix} \vec{a}_1 - \vec{a}_2 & \vec{b}_1 & \vec{b}_2 \end{vmatrix}|}{|\vec{b}_1 \times \vec{b}_2|}$$

ii) If the above two lines are coplanar or intersecting then $\begin{vmatrix} \vec{a}_1 - \vec{a}_2 & \vec{b}_1 & \vec{b}_2 \end{vmatrix} = 0$

iii) Shortest distance between the lines

$$\frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}$$

$$\text{and } \frac{x - x_2}{a_2} = \frac{y - y_2}{b_2} = \frac{z - z_2}{c_2} \text{ is}$$

$$\frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{\sum (b_1 c_2 - b_2 c_1)^2}}$$

W.E-6: The shortest distance between the lines whose vector equations are

$$\vec{r} = (\vec{i} + \vec{j}) + \lambda(2\vec{i} - \vec{j} + \vec{k}) \text{ and}$$

$$\vec{r} = (2\vec{i} + \vec{j} - \vec{k}) + \mu(3\vec{i} - 5\vec{j} + 2\vec{k}) \text{ is}$$

Sol: Compare the given equations with

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1, \vec{r} = \vec{a}_2 + \mu \vec{b}_2$$

$$\therefore \vec{a}_1 = \vec{i} + \vec{j}, \vec{b}_1 = 2\vec{i} - \vec{j} + \vec{k}$$

$$\vec{a}_2 = 2\vec{i} + \vec{j} - \vec{k}, \vec{b}_2 = 3\vec{i} - 5\vec{j} + 2\vec{k}$$

$$\therefore \vec{a}_1 - \vec{a}_2 = -\vec{i} + \vec{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix} = 3\vec{i} - \vec{j} - 7\vec{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{9 + 1 + 49} = \sqrt{59}$$

$$\begin{vmatrix} \vec{a}_1 - \vec{a}_2 & \vec{b}_1 & \vec{b}_2 \end{vmatrix} = \begin{vmatrix} -1 & 0 & 1 \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix} = -10$$

Shortest distance between the skew lines

$$= \frac{|\begin{vmatrix} \vec{a}_1 - \vec{a}_2 & \vec{b}_1 & \vec{b}_2 \end{vmatrix}|}{|\vec{b}_1 \times \vec{b}_2|} = \frac{10}{\sqrt{59}}$$

Distance between parallel lines:

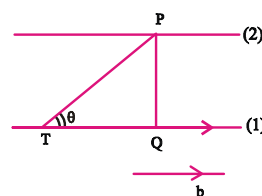
- The distance between the parallel lines

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}, \vec{r} = \vec{a}_2 + \mu \vec{b} \text{ is } \frac{|\vec{b} \times (\vec{a}_1 - \vec{a}_2)|}{|\vec{b}|}$$

Proof: Given parallel lines are:

$$\vec{r} = \vec{a}_1 + \lambda \vec{b} \quad \text{--- (1)}$$

$$\vec{r} = \vec{a}_2 + \lambda \vec{b} \quad \text{--- (2)}$$



Let PQ be the distance between (1) and (2)

let T be a point on (1) with $OT = \vec{a}_1$

$$\text{Let } \vec{OP} = \vec{a}_2$$

Let Q be the projection of P on (1)

Let θ be the angle between PT and $\vec{b}$

$$\vec{b} \times \vec{TP} = (|\vec{b}| |\vec{TP}| \sin \theta) \hat{n} \quad \text{--- (3)}$$

Where $\hat{n}$ is the unit vector perpendicular to the plane of the lines (1) and (2)

$$\vec{TP} = \vec{OP} - \vec{OT} = \vec{a}_2 - \vec{a}_1$$

$$\left[\begin{array}{l} \text{In } \Delta PTQ \\ \sin \theta = \frac{PQ}{PT} \Rightarrow PQ = PT \cdot \sin \theta \end{array} \right]$$

$$\text{From (3)} \quad \vec{b} \times \vec{TP} = (|\vec{b}| |\vec{TP}| \sin \theta) \hat{n}$$

$$\Rightarrow \vec{b} \times (\vec{a}_2 - \vec{a}_1) = |\vec{b}| (PQ) \hat{n}$$

$$\Rightarrow |\vec{b} \times (\vec{a}_2 - \vec{a}_1)| = |\vec{b}| \cdot |\vec{PQ}| \quad (|\hat{n}| = 1)$$

$$\Rightarrow |\vec{PQ}| = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$$

$$\Rightarrow PQ = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$$

$$\Rightarrow PQ = \frac{|\vec{b} \times (\vec{a}_1 - \vec{a}_2)|}{|\vec{b}|}$$

W.E-7 The distance between the parallel lines

given by $\vec{r} = (\vec{i} + 2\vec{j} - 4\vec{k}) + \lambda(2\vec{i} + 3\vec{j} + 6\vec{k})$

and $\vec{r} = (3\vec{i} + 3\vec{j} - 5\vec{k}) + \mu(2\vec{i} + 3\vec{j} + 6\vec{k})$

Sol: Compare the given lines with $\vec{r} = \vec{a}_1 + \lambda\vec{b}$

$$\vec{r} = \vec{a}_2 + \mu\vec{b}$$

$$\vec{a}_1 = \vec{i} + 2\vec{j} - 4\vec{k},$$

$$\vec{a}_2 = 3\vec{i} + 3\vec{j} - 5\vec{k}, \quad \vec{b} = 2\vec{i} + 3\vec{j} + 6\vec{k}$$

$$\text{Distance between the lines} = \frac{|\vec{b} \times (\vec{a}_1 - \vec{a}_2)|}{|\vec{b}|}$$

$$\vec{a}_1 - \vec{a}_2 = -2\vec{i} - \vec{j} + \vec{k}$$

$$\vec{b} \times (\vec{a}_1 - \vec{a}_2) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 6 \\ -2 & -1 & 1 \end{vmatrix} = 9\vec{i} - 14\vec{j} + 4\vec{k}$$

$$|\vec{b} \times (\vec{a}_1 - \vec{a}_2)| = \sqrt{81 + 196 + 16} = \sqrt{293}$$

$$|\vec{b}| = \sqrt{4 + 9 + 36} = 7$$

$$\text{Distance between the lines} = \frac{\sqrt{293}}{7}$$

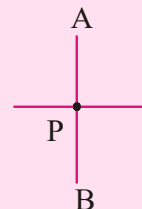
W.E-8 The image of the point (1, 6, 3) in the line

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} \text{ is}$$

Sol. Let A = (1, 6, 3)

Let P be the foot of the perpendicular from A(1, 6, 3) to the given line

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = t$$



then P = (t, 2t+1, 3t+2)

D.r's of $\vec{AP}$ (t-1, 2t-5, 3t-1)

$\vec{AP}$ is perpendicular to the given line

$$\Rightarrow 1(t-1) + 2(2t-5) + 3(3t-1) = 0$$

$$\Rightarrow t = 1, \quad P = (1, 3, 5)$$

Let B be the image of A

$$B = 2P - A = (1, 0, 7)$$

➤ To find the direction of a line with greatest slope:

Let π_1, π_2 be two planes intersecting in a line l_1 then the line of greatest slope in π_1 is the line lying in the plane π_1 and perpendicular to the line l_1 .

Note: Let $\vec{a}, \vec{b}$ be the vectors along the normals to the planes π_1 and π_2 respectively then the vector $\vec{a} \times (\vec{a} \times \vec{b})$ will be along the line of greatest slope in π_1 .

1. The line $m(x-a) = l(z-b)$, $y=c$ is perpendicular to

- 1) X-axis 2) Y-axis
3) Z-axis 4) XY-plane

2. If the lines $x = az + b$, $y = cz + d$; $x = a_1z + b_1$, $y = c_1z + d_1$ are perpendicular then

- 1) $aa_1 + cc_1 + 1 = 0$ 2) $aa_1 + cc_1 - 1 = 0$
3) $aa_1 + bb_1 + 1 = 0$ 4) $aa_1 + dd_1 - 1 = 0$

3. If the equation of line is $x = ay + b$; $z = cy + d$. Then its symmetric form is

- 1) $\frac{x-a}{b} = \frac{y}{1} = \frac{z-c}{d}$ 2) $\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$
3) $\frac{x+a}{b} = \frac{y}{1} = \frac{z+c}{d}$ 4) $\frac{x+b}{a} = \frac{y}{1} = \frac{z+d}{c}$

4. Equation of a line through (α, β, γ) and parallel to Y-axis are

- 1) $\frac{x-\alpha}{1} = \frac{y-\beta}{0} = \frac{z-\gamma}{0}$
2) $\frac{x-\alpha}{0} = \frac{y-\beta}{1} = \frac{z-\gamma}{0}$
3) $\frac{x-\alpha}{0} = \frac{y-\beta}{0} = \frac{z-\gamma}{1}$
4) $\frac{x-\alpha}{1} = \frac{y-\beta}{1} = \frac{z-\gamma}{1}$

5. The equation of the plane containing the line

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_2}{n} \text{ is}$$

a $(x-x_1) + b(y-y_1) + c(z-z_1) = 0$ where

- 1) $ax_1 + by_1 + cz_1 = 0$ 2) $al + bm + cn = 0$
3) $\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$ 4) $lx_1 + my_1 + nz_1 = 0$

6. Equation of a line through (a, b, c) and parallel to z-axis is

1) $\frac{x-a}{1} = \frac{y-b}{0} = \frac{z-c}{0}$

2) $\frac{x-a}{0} = \frac{y-b}{0} = \frac{z-c}{1}$

3) $\frac{x-a}{1} = \frac{y-b}{1} = \frac{z-c}{1}$

4) $\frac{x-a}{0} = \frac{y-b}{1} = \frac{z-c}{0}$

- KEY

- 1) 2 2) 1 3) 2 4) 2 5) 2 6) 2

LEVEL - I

Equation of a line:

1. The equation of the line through $(3,1,2)$ and equally inclined to the axes is

1) $\frac{x-3}{1} = \frac{y-1}{0} = \frac{z-2}{0}$

2) $\frac{x-3}{0} = \frac{y-1}{1} = \frac{z-2}{0}$

3) $\frac{x-3}{0} = \frac{y-1}{0} = \frac{z-2}{1}$

4) $\frac{x-3}{1} = \frac{y-1}{1} = \frac{z-2}{1}$

2. Equation of a line which is parallel to the line $\frac{x-2}{3} = \frac{y+1}{1} = \frac{z-7}{9}$, which passes through the point $(3,0,5)$ is

1) $\frac{x-2}{3} = \frac{y+1}{0} = \frac{z-7}{5}$

2) $\frac{x-2}{0} = \frac{y+1}{1} = \frac{z-7}{4}$

3) $\frac{x-3}{3} = \frac{y-0}{1} = \frac{z-5}{9}$

4) $\frac{x-3}{0} = \frac{y-0}{1} = \frac{z-5}{9}$

3. The equation of the line passing through $(-1, 2, -3)$ and perpendicular to the plane $2x + 3y + z + 5 = 0$ is

$$1) \frac{x-1}{-1} = \frac{y+2}{1} = \frac{z-3}{-1}$$

$$2) \frac{x+1}{-1} = \frac{y-2}{1} = \frac{z+3}{1}$$

$$3) \frac{x+1}{2} = \frac{y-2}{3} = \frac{z+3}{1}$$

$$4) \frac{x+1}{1} = \frac{y-2}{1} = \frac{z+3}{1}$$

4. The cartesian equation of line is

$$\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2} \text{ its vector form is}$$

$$1) \vec{r} = (5\vec{i} + 4\vec{j} + 6\vec{k}) + \lambda(3\vec{i} + 2\vec{j} + 2\vec{k})$$

$$2) \vec{r} = (5\vec{i} - 4\vec{j} + 6\vec{k}) + \lambda(3\vec{i} + 7\vec{j} + 2\vec{k})$$

$$3) \vec{r} = (5\vec{i} + 4\vec{j} + 6\vec{k}) + \lambda(3\vec{i} + 7\vec{j} + 2\vec{k})$$

$$4) \vec{r} = (-5\vec{i} + 4\vec{j} + 6\vec{k}) + \lambda(3\vec{i} + 7\vec{j} + 2\vec{k})$$

Angle between two lines:

5. The value of p so that the lines

$$\frac{1-x}{3} = \frac{7y-14}{2p} = \frac{z-3}{2} \text{ and } \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5} \text{ are}$$

at right angles are

$$1) 70/11 \quad 2) 7/11 \quad 3) 10/7 \quad 4) 17/11$$

6. The angle between the lines

$$\frac{x-1}{2} = \frac{y-2}{1} = \frac{z+3}{2} \text{ and } \frac{x}{1} = \frac{y}{1} = \frac{z}{0} \text{ is}$$

$$1) 0^\circ \quad 2) 30^\circ \quad 3) 45^\circ \quad 4) 90^\circ$$

7. The sine of the angle between the straight

$$\text{line } \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} \text{ and the plane}$$

$$2x - 2y + z = 5 \text{ is}$$

$$1) \frac{\sqrt{5}}{3} \quad 2) \frac{2\sqrt{2}}{5} \quad 3) \frac{1}{5\sqrt{2}} \quad 4) \frac{2\sqrt{3}}{5}$$

8. The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is

$$1) 0^\circ \quad 2) 30^\circ \quad 3) 45^\circ \quad 4) 90^\circ$$

9. The line passing through the points $(5, 1, a)$ and $(3, b, 1)$ crosses the yz -plane at the point

$$\left(0, \frac{17}{2}, \frac{-13}{2}\right) \text{ then (AIEEE - 2008)}$$

$$1) a = 4, b = 6 \quad 2) a = 6, b = 4$$

$$3) a = 8, b = 2 \quad 4) a = 2, b = 8$$

10. The angle between the lines $x = 1, y = 2$ and $y = -1, z = 0$ is

$$1) \frac{\pi}{2} \quad 2) \frac{\pi}{6} \quad 3) \frac{\pi}{3} \quad 4) 0^\circ$$

Coplanar lines:

11. The lines

$$\frac{x-1}{a} = \frac{y-2}{3} = \frac{z-3}{4} ; \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$$

are coplanar. Then $a =$

$$1) 1 \quad 2) 2 \quad 3) -1 \quad 4) -2$$

12. The line $\frac{x+3}{3} = \frac{y-2}{-2} = \frac{z+1}{1}$ and the plane

$$4x + 5y + 3z - 5 = 0 \text{ intersect at a point}$$

$$1) (3, 1, -2) \quad 2) (3, -2, 1) \\ 3) (2, -1, 3) \quad 4) (-1, -2, -3)$$

13. If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and

$$\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1} \text{ are coplanar then } k \text{ can}$$

have

(MAIN-2013)

$$1) \text{ any value} \quad 2) \text{ exactly one value} \\ 3) \text{ exactly two values} \quad 4) \text{ exactly three values}$$

Intersection of lines:

14. A line with d.c's proportional to $(2, 1, 2)$ meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. Then the coordinates of each of the points of intersection are given by

$$1) (3a, 2a, 3a) ; (a, a, 2a)$$

$$2) (3a, 2a, 3a) ; (a, a, a)$$

$$3) (3a, 3a, 3a) ; (a, a, a)$$

$$4) (2a, 3a, 3a) ; (2a, a, a)$$

15. If the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and

$$\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1} \text{ intersect then } k =$$

(JEE MAINS -2012)

- 1) -1 2) 2/9 3) 9/2 4) 0

16. Let L be the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$. If L makes an angle α with the positive X-axis then $\cos \alpha =$ (AIEEE - 2007)

- 1) $\frac{1}{\sqrt{3}}$ 2) $\frac{1}{2}$ 3) 1 4) $\frac{1}{\sqrt{2}}$

LEVEL-I-KEY

- 1) 4 2) 3 3) 3 4) 2 5) 1 6) 3
 7) 3 8) 4 9) 2 10) 1 11) 2 12) 2
 13) 3 14) 2 15) 3 16) 1

LEVEL-II**Equation of a line:**

1. The equation of the line joining $(-2, 1, 3)$ and $(1, 1, 4)$ is

- 1) $\frac{x+2}{3} = \frac{y-1}{0} = \frac{z-3}{1}$
 2) $\frac{x-2}{3} = \frac{y+1}{0} = \frac{z+3}{1}$
 3) $\frac{x+2}{4} = \frac{y+1}{3} = \frac{z-3}{2}$
 4) $\frac{x-3}{1} = \frac{y-1}{1} = \frac{z-2}{1}$

2. Equation of a line passing through $(1, -2, 3)$ and parallel to the plane $2x + 3y + z + 5 = 0$ is

- 1) $\frac{x-1}{-1} = \frac{y+2}{1} = \frac{z-3}{-1}$ 2) $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{1}$
 3) $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z-3}{-1}$ 4) $\frac{x+1}{2} = \frac{y-2}{3} = \frac{z-3}{1}$

3. The equation to the plane which passes through the z-axis and is perpendicular to

the line $\frac{x-1}{\cos \alpha} = \frac{y+2}{\sin \alpha} = \frac{z-3}{0}$ is

- 1) $x \sin \alpha + y \cos \alpha = 0$

2) $x \sin \alpha - y \cos \alpha = 0$

3) $x \cos \alpha + y \sin \alpha = 0$

4) $x \cos \alpha - y \sin \alpha = 0$

4. The vector equation of the line $6x - 2 = 3y + 1 = 2z - 2$ is

1) $\vec{r} = (\vec{i} - \vec{j} + 3\vec{k}) + \lambda(\vec{i} + 2\vec{j} + 3\vec{k})$

2) $\vec{r} = (\vec{i} + 2\vec{j} + 3\vec{k}) + \lambda\left(\frac{1}{3}\vec{i} - \frac{1}{3}\vec{j} + \vec{k}\right)$

3) $\vec{r} = \left(\frac{1}{3}\vec{i} - \frac{1}{3}\vec{j} + \vec{k}\right) + \lambda(\vec{i} + 2\vec{j} + 3\vec{k})$

4) $\vec{r} = (-2\vec{i} + \vec{j} - 2\vec{k}) + \lambda(6\vec{i} + 3\vec{j} + 2\vec{k})$

5. Parametric form of the equation of the line $3x - 6y - 2z - 15 = 0 = 2x + y - 2z - 5$ is

1) $\frac{x-5}{14} = \frac{y}{2} = \frac{z}{15}$ 2) $\frac{x-1}{14} = \frac{y-5}{2} = \frac{z-1}{15}$

3) $\frac{x-3}{14} = \frac{y+1}{2} = \frac{z}{15}$ 4) $\frac{x+5}{14} = \frac{y}{2} = \frac{z}{15}$

Angle between two lines:

6. The value of k if the lines $\frac{x+1}{-3} = \frac{y+2}{2k} = \frac{z-3}{2}$

and $\frac{x-1}{3k} = \frac{y+5}{1} = \frac{z+6}{7}$ may be perpendicular

- 1) 3 2) 2 3) -2 4) 4

7. The line $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is perpendicular to the plane

- 1) $2x + y - 2z = 0$ 2) $3x + 4y + 5z = 7$
 3) $x + y + z = 2$ 4) $2x + 3y + 4z = 0$

8. The angle between the pair of lines

$\vec{r} = (3\vec{i} + 2\vec{j} - 4\vec{k}) + \lambda(\vec{i} + 2\vec{j} + 2\vec{k})$ and

$\vec{r} = (5\vec{i} - 2\vec{j}) + \mu(3\vec{i} + 2\vec{j} + 6\vec{k})$ is

- 1) $\tan^{-1}(19/21)$ 2) $\cos^{-1}(19/21)$

- 3) $\sin^{-1}(19/21)$ 4) $\cos^{-1}(19/20)$
9. If θ is the angle between line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda}z + 4 = 0$ is such that $\sin \theta = \frac{1}{3}$ then the value of λ is
- 1) $\frac{5}{3}$ 2) $\frac{-3}{5}$ 3) $\frac{3}{4}$ 4) $\frac{-4}{3}$
10. The angle between the lines $\frac{x-5}{4} = \frac{y-2}{1} = \frac{z-3}{8}$ and $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ is
- 1) $\cos^{-1}(2/3)$ 2) $\cos^{-1}(1/3)$
3) $\cos^{-1}(4/5)$ 4) $\cos^{-1}(4/3)$
11. The angle between the lines $\frac{x-5}{7} = \frac{y+2}{-5} = \frac{z}{1}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is
- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$
- Coplanar lines:**
12. The equation of the plane containing the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4}$ and perpendicular to the plane $x + 2y + z = 12$ is
- 1) $9x - 2y + 5z + 4 = 0$ 2) $9x - 2y - 5z + 4 = 0$
3) $9x - 2y - 5z - 4 = 0$ 4) $9x + 2y - 5z - 4 = 0$
13. A plane which passes through the point $(3, 2, 0)$ and the line $\frac{x-4}{1} = \frac{y-7}{5} = \frac{z-4}{4}$ is (AIEEE - 2002)
- 1) $x - y + z = 1$ 2) $x + y + z = 5$
3) $x + 2y - 2 = 1$ 4) $2x - y + z = 5$
14. The value of m for which straight line $3x - 2y + z + 3 = 0 = 4x - 3y + 4z + 1$ is parallel to the plane $2x - y + mz - 2 = 0$ is
- 1) -2 2) 8 3) -18 4) 11

Intersection of lines:

15. Let the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lie in the plane $x + 3y - \alpha z + \beta = 0$ then $(\alpha, \beta) =$ (AIEEE - 2009)
- 1) $(-6, 7)$ 2) $(5, -15)$ 3) $(-5, 5)$ 4) $(6, -17)$
16. The d.r.'s of the line given by the planes $x - y + z - 5 = 0, x - 3y - 6 = 0$ are
- 1) $(3, 1, -2)$ 2) $(2, -4, 1)$
3) $(1, -1, 1)$ 4) $(0, 2, 1)$

LEVEL-II-KEY

- 1) 1 2) 1 3) 3 4) 3 5) 3 6) 2
7) 2 8) 2 9) 1 10) 1 11) 3 12) 2
13) 1 14) 1 15) 1 16) 1

LEVEL-III**Distance:**

1. The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along a straight line $x = y = z$ is (AIEEE-2011)
- 1) $3\sqrt{5}$ 2) $10\sqrt{3}$ 3) $5\sqrt{3}$ 4) $3\sqrt{10}$
2. A line passes through two points $A(2, -3, -1)$ and $B(8, -1, 2)$. The coordinates of a point on this line at a distance of 14 units from A are
- 1) $(14, 1, 5)$ 2) $(-10, -7, 7)$
3) $(86, 25, 41)$ 4) $(0, 0, 0)$
3. The distance of the point $P(3, 8, 2)$ from the line $\frac{1}{2}(x-1) = \frac{1}{4}(y-3) = \frac{1}{3}(z-2)$ measured parallel to the plane $3x + 2y - 2z + 15 = 0$ is

- 1) 7 2) 9 3) $\sqrt{7}$ 4) 49

Point of intersection:

4. The line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the curve $xy = c^2, z = 0$ if $c =$
- 1) ± 1 2) $\pm\sqrt{3}$ 3) $\pm\sqrt{5}$ 4) $\pm\sqrt{7}$
5. The point of intersection of the line $\frac{x-3}{3} = \frac{2-y}{4} = \frac{z+1}{1}$ and planes $2x + 4y + 3z + 3 = 0, x + 2y + 3z = 0$ is

- 1) (9,6,1) 2) (-9,6,1)
 3) (9,-6,1) 4) (-9,-6,-1)

Shortest distance:

6. The shortest distance between the lines

$$\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} ; \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \text{ is}$$

1) 2 2) 1 3) 3 4) 0

7. The shortest distance between the lines

$$\vec{r} = (2\vec{i} - \vec{j} - \vec{k}) + \lambda(2\vec{i} + \vec{j} + 2\vec{k}) \text{ and}$$

$$\vec{r} = (\vec{i} + 2\vec{j} + \vec{k}) + \mu(\vec{i} - \vec{j} + \vec{k}) \text{ is}$$

- 1) $3/\sqrt{2}$ 2) $3\sqrt{2}$ 3) $2/\sqrt{3}$ 4) $\sqrt{2}/3$

Distance between parallel lines:

8. The distance between the parallel lines

$$\vec{r} = 2\vec{i} + 3\vec{j} - \vec{k} + \lambda(\vec{i} - \vec{j} + 2\vec{k}) \text{ and}$$

$$\vec{r} = -3\vec{i} + 4\vec{j} + \vec{k} + \mu(\vec{i} - \vec{j} + 2\vec{k}) \text{ is}$$

- 1) $2\sqrt{\frac{22}{3}}$ 2) $\sqrt{\frac{175}{6}}$ 3) $\frac{14}{\sqrt{6}}$ 4) 7

Foot and Image:

9. The reflection of the point A (1, 0, 0) in the

$$\text{line } \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8} \text{ is}$$

- 1) (3, -4, -2) 2) (5, -8, -4)
 3) (1, -1, -10) 4) (2, -3, 8)

10. The foot of the perpendicular from (a,b,c) on the line $x = y = z$ is the point (r,r,r) where

- 1) $r = a + b + c$ 2) $r = 3(a + b + c)$
 3) $3r = a + b + c$ 4) $r = 4(a + b + c)$

Length of the perpendicular:

11. The perpendicular distance of the point

$$(2,4,-1) \text{ from line } \frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9} \text{ is}$$

- 1) 3 2) 5 3) 7 4) 9

12. The length of perpendicular drawn from the

$$\text{point } (3,-1,11) \text{ to the line } \frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$$

is (AIEEE-2011)

- 1) $\sqrt{66}$ 2) $\sqrt{29}$ 3) $\sqrt{33}$ 4) $\sqrt{53}$

LEVEL-III-KEY

- 1) 2 2) 1 3) 1 4) 3 5) 3 6) 4
 7) 1 8) 1 9) 2 10) 3 11) 3 12) 4

LEVEL-IV

Distance:

1. The distance of the point (3, -4, 5) from the plane $2x + 5y - 6z = 16$ measured along a line with d.cs, proportional to (2, 1, -2) is

- 1) $\frac{60}{7}$ 2) $\frac{50}{7}$ 3) $\frac{30}{7}$ 4) $\frac{20}{7}$

2. The distance of the point (1,0,-3) from the plane $x - y - z = 9$ measured parallel to the line $\frac{x-2}{2} = \frac{y+2}{3} = \frac{z-6}{-6}$ is

- 1) 6 2) 7 3) 8 4) 9

Point of intersection:

3. A line with d.r's (2,7,-5) is drawn to intersect the lines $\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$ and

$$\frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4} \text{ at P and Q respectively.}$$

Length of PQ is

- 1) $\sqrt{78}$ 2) $\sqrt{77}$ 3) $\sqrt{54}$ 4) $\sqrt{74}$

4. If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and

$$\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2} \text{ intersect at a point then the integer k is equal to}$$

- 1) 5 2) 2 3) -2 4) -5

Shortest distance:

5. The shortest distance between the lines

$$\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-1}{2} ; \frac{x-4}{4} = \frac{y-5}{5} = \frac{z-2}{3} \text{ is}$$

- 1) $\frac{1}{\sqrt{3}}$ 2) $\frac{1}{\sqrt{6}}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{5}{\sqrt{6}}$

6. The shortest distance between the lines
 $\vec{r} = (1-t)\vec{i} + (t-2)\vec{j} + (3-2t)\vec{k}$ and
 $\vec{r} = (s+1)\vec{i} + (2s-1)\vec{j} - (2s+1)\vec{k}$ is

- 1) $8/\sqrt{17}$ 2) $8/\sqrt{493}$
 3) $8/\sqrt{29}$ 4) $16\sqrt{29}$

7. The length of the shortest distance between the lines $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$ and

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \text{ is}$$

- 1) $4\sqrt{29}$ 2) $3\sqrt{29}$ 3) $\sqrt{29}$ 4) $2\sqrt{29}$

Distance between parallel lines:

8. The distance between the parallel lines
 $\vec{r} = (\vec{i} - \vec{j}) + \lambda(3\vec{i} + \vec{j} - \vec{k})$ and
 $\vec{r} = (\vec{j} - \vec{k}) + \mu(2\vec{i} + \vec{j} - \vec{k})$ is

- 1) 6 2) 7 3) $\sqrt{\frac{35}{6}}$ 4) $\sqrt{\frac{33}{6}}$

Foot and Image:

9. Foot of the perpendicular drawn from the point A (1,0,3) to the join of the points B (4,7,1) and C (3,5,3) is

- 1) (1,2,-2) 2) $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$
 3) $\left(\frac{1}{3}, \frac{2}{3}, \frac{-2}{3}\right)$ 4) $\left(\frac{-5}{3}, \frac{-7}{3}, \frac{-17}{3}\right)$

10. The image of the point (3, -1, 11) w.r.t the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is

- 1) (2, 5, 7) 2) (1, 11, 3)
 3) (0, 0, 0) 4) (0, 2, 3)

Length of the perpendicular:

11. The length of the perpendicular from the point (1, 2, 3) to the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$ is

- 1) 7 2) $\sqrt{48}$ 3) 8 4) 9

12. The length of the perpendicular from the point (-1, 3, 9) to the line $\frac{x-13}{5} = \frac{y+8}{-8} = \frac{z-31}{1}$ is

- 1) 21 2) 22 3) 20 4) $\sqrt{439}$

LEVEL-IV-KEY

- 1) 1 2) 2 3) 1 4) 4 5) 2 6) 3
 7) 4 8) 3 9) 2 10) 2 11) 1 12) 1

LEVEL-V

1. A plane mirror is placed at the origin so that the direction ratios of its normal are (1, -1, 1). A ray of light, coming along the positive direction of the x-axis strikes the mirror. Then the direction ratios of the reflected ray are

- 1) $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}$ 2) $\frac{-1}{3}, \frac{2}{3}, \frac{2}{3}$
 3) $\frac{-1}{3}, \frac{-2}{3}, \frac{-2}{3}$ 4) $\frac{-1}{3}, \frac{-2}{3}, \frac{2}{3}$

2. Equation of the perpendicular line from (3, -1, 11) to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is

- 1) $\frac{x-3}{1} = \frac{y+1}{-6} = \frac{z-11}{4}$ 2) $\frac{x-3}{2} = \frac{y+1}{5} = \frac{z-11}{7}$
 3) $\frac{x-3}{1} = \frac{y+1}{11} = \frac{z-11}{3}$ 4) $\frac{x-3}{1} = \frac{y+1}{6} = \frac{z-11}{4}$

3. The equation of line of shortest distance between the lines $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$;

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \text{ is}$$

- 1) $\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$ 2) $\frac{x+3}{2} = \frac{y+5}{3} = \frac{z-7}{4}$
 3) $\frac{x+3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$ 4) $\frac{x-3}{2} = \frac{y-5}{3} = \frac{z+7}{4}$

4. Let P(3, 2, 6) be a point in space and Q be a point on the line

$\vec{r} = (\vec{i} - \vec{j} + 2\vec{k}) + \mu(-3\vec{i} + \vec{j} + 5\vec{k})$ then the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is (IIT-2009)

- 1) $\frac{1}{4}$ 2) $-\frac{1}{4}$ 3) $\frac{1}{8}$ 4) $-\frac{1}{8}$

5. If the angle between the line $x = \frac{y-1}{2} = \frac{z-3}{\lambda}$ and the plane $x + 2y + 3z = 4$ is

$\cos^{-1}(\sqrt{5/14})$ then $\lambda =$ (AIEEE-2011)

- 1) $\frac{3}{2}$ 2) $\frac{5}{3}$ 3) $\frac{2}{3}$ 4) $\frac{2}{5}$

6. If lines $x = y = z$ and $x = y/2 = z/3$ and third line passing through $(1, 1, 1)$ form a triangle of area $\sqrt{6}$ units, then point of intersection of third line with second line will be

- 1) $(1, 2, 3)$ 2) $(2, 4, 6)$
3) $(\frac{4}{3}, \frac{8}{3}, \frac{12}{3})$ 4) $(2, 1, 3)$

7. The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane $2x - y + z + 3 = 0$ is the line (MAINS-2014)

- 1) $\frac{x-3}{3} = \frac{y+5}{1} = \frac{z-2}{-5}$
2) $\frac{x-3}{-3} = \frac{y+5}{-1} = \frac{z-2}{5}$
3) $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$
4) $\frac{x+3}{-3} = \frac{y-5}{-1} = \frac{z+2}{5}$

8. For the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$, which one of the following is incorrect?

1) it lies in the plane $x - 2y + z = 0$

2) it is same as line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$

3) it passes through $(2, 3, 5)$

4) it is parallel to the plane $x - 2y + z = 6$

9. The equation of a plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and at greatest distance from point $(0, 0, 0)$ is

1) $4x + 3y + 5z = 25$ 2) $4x + 3y + 5z = 50$

3) $4x + 3y + 5z = 49$ 4) $x + 7y - 5z = 2$

10. The equation of the plane which passes through the z-axis and is perpendicular to the line $\frac{x-a}{\cos \theta} = \frac{y+2}{\sin \theta} = \frac{z-3}{0}$ is

1) $x + y \tan \theta = 0$ 2) $y + x \tan \theta = 0$

3) $x \cos \theta - y \sin \theta = 0$

4) $x \sin \theta - y \cos \theta = 0$

11. If the straight lines $x = 1 + s$, $y = -3 - \lambda s$, $z = 1 + \lambda s$ and $x = \frac{t}{2}$, $y = 1 + t$, $z = 2 - t$ with parameters 's' and 't' respectively are coplanar then $\lambda =$

- 1) -2 2) 0 3) $-\frac{1}{2}$ 4) -1

12. The projection of the line $\frac{x+1}{-1} = \frac{y}{2} = \frac{z-1}{3}$ in the plane $x - 2y + z = 6$ is the line of intersection of this plane with the plane is

1) $2x+y+2=0$ 2) $3x+y-z=2$

3) $2x-3y+8z=3$ 4) $x+y-z=1$

LEVEL-V-KEY

1) 4 2) 1 3) 1 4) 1 5) 3 6) 2

7) 3 8) 3 9) 2 10) 1 11) 1 12) 1

LEVEL-VI**ASSERTION & REASON:**

The following questions consist of two statements one labelled as , Assertion (A) and the other labelled as Reason (R) You are to examine these two statements carefully and decide if the Assertion (A) and the Reason (R) are individually true and if so, whether the Reason (R) is the correct explanation for the given Assertion (A) select your answer to these items using the codes given below and then select the correct option

Codes:

(A) Both A and R are individually true and R is the correct explanation of A

(B) Both A and R individually true but R is not the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

1. Assertion (A) : The angle between the lines

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{0} \text{ and } \frac{x-1}{2} = \frac{y-2}{1} = \frac{z+3}{2} \text{ is } \frac{\pi}{4}$$

Reason (R): The acute angle between the

lines given by $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ and

$$\frac{x-x_2}{p} = \frac{y-y_2}{q} = \frac{z-z_2}{r} \text{ then}$$

$$\cos \theta = \frac{pl + qm + rn}{\sqrt{\sum l^2} \sqrt{\sum p^2}}$$

1) A 2) B 3) C 4) D

2. Statement-1: The point $A(1,0,7)$ is the mirror image of the point $B(1,6,3)$ in the

line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

Statement-2: The line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

bisects the line segment joining $A(1,0,7)$

and $B(1,6,3)$ [AIEEE-2011]

1) Statement-1 is true, statement -2 is true; statement - 2 is not correct explanation for statement-1

2) Statement-1 is true, statement-2 is false

3) Statement -1 is false, statement-2 is true

4) Statement-1 is true, statement -2 is true, statement -2 is a correct explanation for statement-1.

LEVEL-IV-KEY

1) 1 2) 1

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